

Non-Abelian Gauge Groups and $SO(n)$ Yang–Mills Theory

Road map

This chapter develops $SO(n)$ gauge theory from first principles. We begin with n free real scalar fields and separate spacetime symmetry from internal symmetry. We then derive the Lie algebra $\mathfrak{so}(n)$, count its generators, compute all Noether currents, and explain the passage from a finite-dimensional global structure group to the infinite-dimensional group of spacetime-dependent gauge transformations. The gauge principle forces the introduction of a connection A_μ and the covariant derivative D_μ . Their transformation laws are derived rather than guessed. Finally, the commutator of covariant derivatives produces the non-Abelian field strength, whose invariant quadratic term is the Yang–Mills Lagrangian. The comparison table, the geometric fibre picture, and the elementary interaction vertex in the handwritten lecture are all reproduced below.

12.1 A free multiplet of real scalar fields

12.1.1 Fields, spacetime, and the action

Let M be four-dimensional Minkowski spacetime and let

$$\varphi_i : M \longrightarrow \mathbb{R}, \quad x \longmapsto \varphi_i(x), \quad i = 1, \dots, n,$$

be n real scalar fields. The index i labels the fields; it is an *internal* index and is completely passive under a change of spacetime coordinates. We use

$$x = x^\mu = (x^0, x^1, x^2, x^3) = (ct, x^1, x^2, x^3), \quad \mu = 0, 1, 2, 3,$$

and, unless units are being displayed explicitly, set $c = 1$. With the mostly-minus Minkowski metric $\eta = \text{diag}(1, -1, -1, -1)$, indices are raised by $\partial^\mu = \eta^{\mu\nu} \partial_\nu$.

The free action in the notes is

$$\begin{aligned} S[\varphi_1, \dots, \varphi_n] &= \int_M d^4x \sum_{i=1}^n \left[\frac{1}{2} \sum_{\mu=0}^3 \partial_\mu \varphi_i(x) \partial^\mu \varphi_i(x) - \frac{1}{2} m^2 \varphi_i(x)^2 \right] \\ &= \int_M d^4x \mathcal{L}(\Phi, \partial_\mu \Phi), \end{aligned} \quad (12.1)$$

where the field multiplet is the column vector

$$\Phi(x) = \begin{pmatrix} \varphi_1(x) \\ \varphi_2(x) \\ \vdots \\ \varphi_n(x) \end{pmatrix} \in \mathbb{R}^n.$$

In vector notation the Lagrangian density becomes

$$\mathcal{L}_{\text{mat}} = \frac{1}{2} \partial_\mu \Phi^T \partial^\mu \Phi - \frac{1}{2} m^2 \Phi^T \Phi. \quad (12.2)$$

It is quadratic in both the fields and their first derivatives. It therefore describes a free theory: the equations of motion will be linear and their solutions may be superposed.

12.1.2 Lorentz invariance and the wave equations

Under a Lorentz transformation,

$$x'^\mu = \Lambda^\mu_\nu x^\nu, \quad \Lambda^T \eta \Lambda = \eta.$$

Because every φ_i is a scalar, $\varphi'_i(x') = \varphi_i(x)$, whereas the derivatives transform as a covector:

$$\partial'_\mu = (\Lambda^{-1})^\nu_\mu \partial_\nu.$$

Consequently,

$$\partial'_\mu \varphi'_i(x') \partial'^\mu \varphi'_i(x') = \partial_\mu \varphi_i(x) \partial^\mu \varphi_i(x),$$

and $d^4x' = d^4x$ for a proper Lorentz transformation. Thus the action is Lorentz invariant. Noether's theorem associates the Lorentz rotations and boosts with conservation of relativistic angular momentum.

For each field, the Euler–Lagrange equation is

$$0 = \frac{\partial \mathcal{L}}{\partial \varphi_i} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi_i)}.$$

The two derivatives appearing here are

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} = -m^2 \varphi_i, \quad \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi_i)} = \partial^\mu \varphi_i.$$

Substitution gives

$$\begin{aligned} 0 &= -m^2\varphi_i - \partial_\mu\partial^\mu\varphi_i, \\ (\partial_\mu\partial^\mu + m^2)\varphi_i &= 0. \end{aligned}$$

Writing $\square = \partial_\mu\partial^\mu$, all n fields satisfy the Klein–Gordon equation

$$(\square + m^2)\varphi_i = 0, \quad i = 1, \dots, n. \quad (12.3)$$

For a plane-wave mode

$$\varphi_i(x) = c_i e^{-ip_\mu x^\mu},$$

one has $\partial_\mu\varphi_i = -ip_\mu\varphi_i$, so (12.3) becomes

$$(-p_\mu p^\mu + m^2)\varphi_i = 0.$$

A nonzero mode therefore obeys the mass-shell condition

$$p_\mu p^\mu = m^2, \quad E^2 = \mathbf{p}^2 + m^2.$$

The general real solution is a linear superposition of positive- and negative-frequency modes $e^{-iEt+i\mathbf{p}\cdot\mathbf{x}}$ and $e^{+iEt-i\mathbf{p}\cdot\mathbf{x}}$, with $E = \sqrt{\mathbf{p}^2 + m^2}$, combined so that the result is real.

12.2 The internal structure group $SO(n)$

12.2.1 The fundamental action on the multiplet

The theory has a symmetry unrelated to spacetime: one may rotate the n field components among themselves. For

$$R \in SO(n) = \{R \in M_n(\mathbb{R}) : R^T R = I, \det R = 1\},$$

the defining, or fundamental, representation is

$$D : SO(n) \longrightarrow GL(n, \mathbb{R}), \quad D(R) = R.$$

It is a representation because

$$D(R_1 R_2) = R_1 R_2 = D(R_1) D(R_2).$$

The action on fields is pointwise:

$$\Phi(x) \longmapsto D(R)\Phi(x) = R\Phi(x), \quad \varphi_i(x) \longmapsto \sum_{j=1}^n R_{ij}\varphi_j(x). \quad (12.4)$$

The spacetime point x is unchanged. This is why $SO(n)$ is called an *internal* structure group here.

For a *global* transformation, R is independent of x . Hence

$$\partial_\mu(R\Phi) = R\partial_\mu\Phi.$$

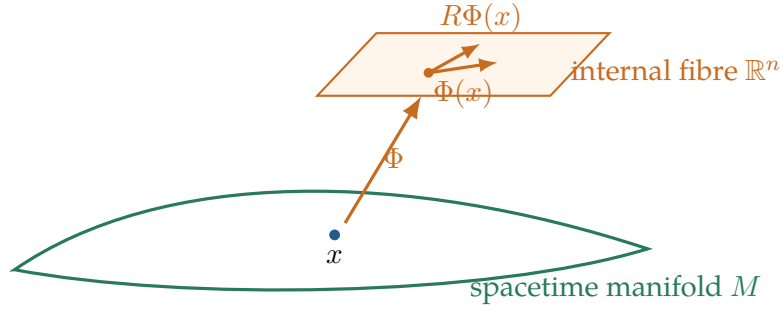


图 12.1: An internal rotation changes the vector in the fibre over x ; it does not move the base point $x \in M$.

Using $R^T R = I$,

$$\begin{aligned} (R\Phi)^T (R\Phi) &= \Phi^T R^T R \Phi = \Phi^T \Phi, \\ \partial_\mu (R\Phi)^T \partial^\mu (R\Phi) &= \partial_\mu \Phi^T R^T R \partial^\mu \Phi = \partial_\mu \Phi^T \partial^\mu \Phi. \end{aligned}$$

Both terms in (12.2) are invariant. Therefore the free action has global $SO(n)$ internal symmetry.

12.2.2 Infinitesimal rotations and the Lie algebra

Let $R(\varepsilon)$ be a differentiable curve in $SO(n)$ through the identity:

$$R(0) = I, \quad R(\varepsilon) = I + \varepsilon T + O(\varepsilon^2).$$

Orthogonality for every ε requires

$$\begin{aligned} I &= R(\varepsilon)^T R(\varepsilon) \\ &= (I + \varepsilon T^T + O(\varepsilon^2))(I + \varepsilon T + O(\varepsilon^2)) \\ &= I + \varepsilon(T^T + T) + O(\varepsilon^2). \end{aligned}$$

The coefficient of ε must vanish, so

$$T^T = -T.$$

Thus the generators of $SO(n)$ are real skew-symmetric matrices.

Definition 12.1: The Lie algebra of $SO(n)$

$$\mathfrak{so}(n) = \{T \in M_n(\mathbb{R}) : T^T = -T\}.$$

It is the tangent space at the identity of $SO(n)$, equipped with the matrix commutator $[T_1, T_2] = T_1 T_2 - T_2 T_1$.

Skew-symmetry fixes every diagonal entry to zero and fixes the lower triangle from the upper triangle:

$$T_{ii} = 0, \quad T_{ji} = -T_{ij}.$$

There is one independent entry for each pair $1 \leq i < j \leq n$. Therefore

$$\dim \mathfrak{so}(n) = \binom{n}{2} = \frac{1}{2}n(n-1). \quad (12.5)$$

Geometrically, each generator produces a rotation in one of the coordinate two-planes.

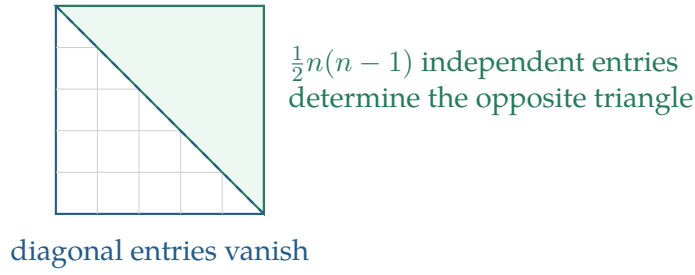


图 12.2: The independent data of a skew-symmetric matrix occupy one strict triangular half.

The Lie algebra is closed under real linear combinations. It is also closed under the commutator. Indeed, if $T_1^T = -T_1$ and $T_2^T = -T_2$, then

$$\begin{aligned}
 [T_1, T_2]^T &= (T_1 T_2 - T_2 T_1)^T \\
 &= T_2^T T_1^T - T_1^T T_2^T \\
 &= (-T_2)(-T_1) - (-T_1)(-T_2) \\
 &= T_2 T_1 - T_1 T_2 \\
 &= -[T_1, T_2].
 \end{aligned}$$

Hence $[T_1, T_2] \in \mathfrak{so}(n)$.

Choose a basis

$$T^1, \dots, T^d, \quad d = \frac{1}{2}n(n-1),$$

of $\mathfrak{so}(n)$. The infinitesimal rotation generated by T^a acts as

$$\begin{aligned}
 \Phi &\mapsto (I + \varepsilon T^a)\Phi + O(\varepsilon^2), \\
 \varphi_i(x) &\mapsto \varphi_i(x) + \varepsilon \sum_{j=1}^n T_{ij}^a \varphi_j(x) + O(\varepsilon^2), \\
 \delta_a \varphi_i(x) &= \varepsilon \sum_{j=1}^n T_{ij}^a \varphi_j(x).
 \end{aligned} \tag{12.6}$$

Thus there are exactly d linearly independent infinitesimal internal rotations.

12.3 Noether currents for the global internal symmetry

12.3.1 Direct check of infinitesimal invariance

For the variation (12.6),

$$\delta_a(\partial_\mu \varphi_i) = \partial_\mu(\delta_a \varphi_i) = \varepsilon \sum_j T_{ij}^a \partial_\mu \varphi_j,$$

because ε and T^a are constant. The first variation of the Lagrangian is

$$\begin{aligned}\delta_a \mathcal{L} &= \sum_i \left[\frac{\partial \mathcal{L}}{\partial \varphi_i} \delta_a \varphi_i + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi_i)} \delta_a (\partial_\mu \varphi_i) \right] \\ &= -\varepsilon m^2 \sum_{i,j} \varphi_i T_{ij}^a \varphi_j + \varepsilon \sum_{i,j} \partial^\mu \varphi_i T_{ij}^a \partial_\mu \varphi_j.\end{aligned}$$

Each quadratic contraction vanishes. For any real vector v ,

$$v^T T^a v = (v^T T^a v)^T = v^T (T^a)^T v = -v^T T^a v,$$

so $v^T T^a v = 0$. Apply this once to $v = \Phi$ and once, for each μ , to $v = \partial_\mu \Phi$. Therefore

$$\delta_a \mathcal{L} = 0.$$

12.3.2 The $\frac{1}{2}n(n-1)$ conserved currents

Noether's first theorem gives one conserved current for every generator. Omitting the common infinitesimal parameter ε , the current is

$$\begin{aligned}j^{\mu a} &= \sum_i \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi_i)} \sum_j T_{ij}^a \varphi_j \\ &= \sum_{i,j} \partial^\mu \varphi_i T_{ij}^a \varphi_j = \partial^\mu \Phi^T T^a \Phi.\end{aligned}\tag{12.7}$$

On solutions of the field equations,

$$\partial_\mu j^{\mu a} = 0, \quad a = 1, \dots, \frac{1}{2}n(n-1).\tag{12.8}$$

This can also be checked directly:

$$\begin{aligned}\partial_\mu j^{\mu a} &= (\square \Phi)^T T^a \Phi + \partial_\mu \Phi^T T^a \partial^\mu \Phi \\ &= (-m^2 \Phi)^T T^a \Phi + 0 = 0.\end{aligned}$$

The mathematical generators T^a are real and skew-symmetric. Physics often uses instead the Hermitian generators

$$H^a = iT^a, \quad (H^a)^\dagger = H^a.$$

Multiplying (12.7) by i gives the equally conserved current

$$\hat{j}^{\mu a} = \partial^\mu \Phi^T iT^a \Phi = \partial^\mu \Phi^T H^a \Phi, \quad \partial_\mu \hat{j}^{\mu a} = 0.\tag{12.9}$$

This is the convention in which the generators are represented by Hermitian observables. It amounts to working in the complexification of $\mathfrak{so}(n)$; no physical conservation law is changed.

12.4 From a global symmetry to a gauge symmetry

12.4.1 Structure group versus gauge group

We now promote the constant internal rotation to a spacetime-dependent one:

$$R \longrightarrow R(x).$$

The corresponding gauge group is

$$\mathcal{G} = C^\infty(M, SO(n)),$$

with pointwise multiplication $(R_1 R_2)(x) = R_1(x) R_2(x)$. The distinction made in the lecture is summarized in the following table.

Structure group $G = SO(n)$	Gauge group $\mathcal{G} = C^\infty(M, SO(n))$
A Lie group acting on the n -dimensional internal field space: $\Phi \mapsto R\Phi$.	The group of smooth functions from spacetime to G : $\Phi(x) \mapsto R(x)\Phi(x)$.
A finite number $d = \frac{1}{2}n(n-1)$ of independent generators T^a .	An independent infinitesimal coefficient may be chosen at every spacetime point: $T^a \mapsto \alpha^a(x)T^a$.
Finite-dimensional Lie group.	Infinite-dimensional Lie group.

表 12.1: Global internal rotations and local gauge transformations.

Demanding invariance under arbitrary $R(x)$ is much more restrictive than demanding invariance under a constant R . Starting with a free theory, this demand necessarily introduces additional fields and interactions. This construction is the *gauge principle*: it is a systematic recipe for building interacting field theories from a chosen internal structure group. In this sense, fundamental interactions arise from local symmetry requirements.

Quantum electrodynamics and the Standard Model are gauge theories. Their structure group is

$$U(1) \times SU(2) \times SU(3),$$

where the three factors describe, respectively, the electromagnetic, weak, and strong gauge sectors. Nothing in the construction is restricted to these examples: any suitable Lie group can be used. Non-Abelian groups such as $SU(2)$, $SU(3)$, and $SO(n)$ are particularly interesting because their gauge particles interact with one another.

A condensed-matter analogy

For a classical Heisenberg model on a lattice, one may write

$$H = \sum_{\langle x, x' \rangle} J_{xx'} \mathbf{S}_x \cdot \mathbf{S}_{x'}, \quad \mathbf{S}_x^2 = \text{constant}.$$

Replacing a normalized spin by components $\mathbf{S}_x = (\varphi_{1x}, \dots, \varphi_{nx})$ gives the constraint

$$\sum_i \varphi_{ix}^2 = 1.$$

In a continuum limit these become fields $\varphi_i(x)$. A description in which $\Phi(x)$ is replaced by $R(x)\Phi(x)$ without changing the physical configuration contains a local redundancy. This is a useful elementary picture of how gauge structure can arise naturally; the local transformation is a redundancy of description, not an operation producing a different physical state.

12.4.2 Why the ordinary derivative fails

The global transformation law (12.4) becomes

$$\Phi'(x) = R(x)\Phi(x), \quad R(x) \in SO(n).$$

The mass term remains invariant pointwise:

$$\Phi'^T \Phi' = \Phi^T R^T R \Phi = \Phi^T \Phi.$$

The kinetic term does not. The product rule gives

$$\partial_\mu \Phi' = (\partial_\mu R)\Phi + R \partial_\mu \Phi \neq R \partial_\mu \Phi \quad (12.10)$$

unless R is constant. The extra term $(\partial_\mu R)\Phi$ compares internal vectors using frames that vary from point to point. To remove it, we need a rule for parallel comparison: a connection.

12.5 The covariant derivative and the gauge field

12.5.1 Minimal substitution

Introduce a matrix-valued field $A_\mu(x)$, the *gauge field*, and a real coupling constant g . In the Hermitian-generator convention define

$$D_\mu = \partial_\mu - igA_\mu(x). \quad (12.11)$$

Replacing ∂_μ with D_μ is the *minimal substitution*. The locally coupled matter Lagrangian is

$$\mathcal{L}_{\text{matter}} = \frac{1}{2}(D_\mu \Phi)^T D^\mu \Phi - \frac{1}{2}m^2 \Phi^T \Phi. \quad (12.12)$$

We require the covariant derivative of a field to transform in the same way as the field itself:

$$D'_\mu \Phi' = R D_\mu \Phi. \quad (12.13)$$

Then

$$(D'_\mu \Phi')^T D'^\mu \Phi' = (R D_\mu \Phi)^T (R D^\mu \Phi) = (D_\mu \Phi)^T D^\mu \Phi,$$

so (12.12) is locally gauge invariant.

12.5.2 Derivation of the connection transformation law

We now determine A'_μ from (12.13). The right-hand side is

$$R D_\mu \Phi = R \partial_\mu \Phi - ig R A_\mu \Phi.$$

The left-hand side is

$$\begin{aligned} D'_\mu \Phi' &= (\partial_\mu - igA'_\mu)(R\Phi) \\ &= (\partial_\mu R)\Phi + R\partial_\mu \Phi - igA'_\mu R\Phi. \end{aligned}$$

After cancelling $R\partial_\mu \Phi$, equality for every Φ gives

$$-igRA_\mu = \partial_\mu R - igA'_\mu R.$$

Equivalently,

$$RA_\mu = \frac{i}{g}\partial_\mu R + A'_\mu R,$$

and multiplication on the right by R^{-1} yields

$$A'_\mu = RA_\mu R^{-1} - \frac{i}{g}(\partial_\mu R)R^{-1}. \quad (12.14)$$

Differentiate $RR^{-1} = I$:

$$0 = \partial_\mu(RR^{-1}) = (\partial_\mu R)R^{-1} + R\partial_\mu R^{-1}.$$

Thus $-(\partial_\mu R)R^{-1} = R\partial_\mu R^{-1}$, and the equivalent form recorded in the notes is

$$\boxed{A'_\mu = RA_\mu R^{-1} + \frac{i}{g}R\partial_\mu R^{-1}.} \quad (12.15)$$

The inhomogeneous derivative term is what distinguishes a connection from an ordinary matter field.

There is also a concise operator derivation. Equation (12.13) implies, for every field,

$$D'_\mu R\Phi = RD_\mu \Phi.$$

Therefore $D'_\mu R = RD_\mu$, and hence

$$\boxed{D'_\mu = RD_\mu R^{-1}.} \quad (12.16)$$

Expanding this identity reproduces (12.15).

12.5.3 Why the transformed field remains Lie-algebra valued

Let T^a be real skew-symmetric generators and put

$$H^a = iT^a.$$

The matrices H^a are Hermitian, and we expand the gauge field as

$$A_\mu(x) = \sum_{a=1}^d A_\mu^a(x) H^a, \quad d = \frac{1}{2}n(n-1). \quad (12.17)$$

The real functions $A_\mu^a(x)$ are the component gauge fields. This is the physics convention corresponding to the complexified Lie algebra. In the real skew convention one may instead define

$$B_\mu = -igA_\mu \in \mathfrak{so}(n), \quad D_\mu = \partial_\mu + B_\mu.$$

Then

$$B'_\mu = RB_\mu R^{-1} + R\partial_\mu R^{-1},$$

and both terms are manifestly real and skew-symmetric.

For completeness, this closure can be checked directly. First,

$$(RB_\mu R^{-1})^T = (R^{-1})^T B_\mu^T R^T = R(-B_\mu)R^{-1} = -RB_\mu R^{-1}.$$

Second, differentiating $R^T R = I$ gives

$$(\partial_\mu R^T)R + R^T \partial_\mu R = 0.$$

It follows that

$$(R\partial_\mu R^{-1})^T = -R\partial_\mu R^{-1}.$$

Thus $B'_\mu \in \mathfrak{so}(n)$, or equivalently A'_μ remains in the Hermitian-generator version of the same Lie algebra.

12.6 Interactions forced by the gauge principle

12.6.1 Expansion of the matter Lagrangian

Expanding the covariant derivatives in (12.12) gives

$$\begin{aligned} (D_\mu \Phi)^T D^\mu \Phi &= (\partial_\mu \Phi - igA_\mu \Phi)^T (\partial^\mu \Phi - igA^\mu \Phi) \\ &= \partial_\mu \Phi^T \partial^\mu \Phi - ig \Phi^T A_\mu^T \partial^\mu \Phi - ig \partial_\mu \Phi^T A^\mu \Phi - g^2 \Phi^T A_\mu^T A^\mu \Phi. \end{aligned}$$

Therefore

$$\mathcal{L}_{\text{matter}} = \mathcal{L}_{\text{free}} + \mathcal{L}_{\text{int}},$$

where

$$\begin{aligned} \mathcal{L}_{\text{free}} &= \frac{1}{2} \partial_\mu \Phi^T \partial^\mu \Phi - \frac{1}{2} m^2 \Phi^T \Phi, \\ \mathcal{L}_{\text{int}} &= \frac{1}{2} [-ig \Phi^T A_\mu^T \partial^\mu \Phi - ig \partial_\mu \Phi^T A^\mu \Phi - g^2 \Phi^T A_\mu^T A^\mu \Phi]. \end{aligned} \quad (12.18)$$

The first two terms are cubic in the fields and first order in g ; the last is quartic and second order in g . Thus the attempt to make the free matter theory locally invariant has automatically generated interactions.

After quantization, the cubic terms describe elementary processes in which a matter quantum changes its internal component while emitting or absorbing a gauge quantum. A schematic first-order vertex from the notes is:

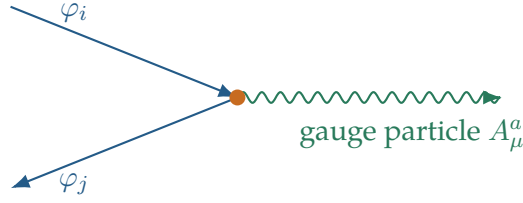


图 12.3: A schematic matter–matter–gauge-field interaction generated at first order in g .

12.7 Dynamics of the gauge field

12.7.1 Why the Maxwell-type first guess is incomplete

The field A_μ needs its own dynamical term. By analogy with electromagnetism, a first guess would be

$$f_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a, \quad -\frac{1}{4} \sum_a f_{\mu\nu}^a f^{\mu\nu a}.$$

This is Lorentz invariant, but for a non-Abelian group it is not gauge invariant. The transformation (12.15) contains both a conjugation and a derivative term, and differentiating it produces additional terms that do not cancel in the Abelian expression.

The correct replacement is to use covariant derivatives:

$$F_{\mu\nu} = D_\mu A_\nu - D_\nu A_\mu. \quad (12.19)$$

Expanding $D_\mu = \partial_\mu - igA_\mu$ yields

$$\begin{aligned} F_{\mu\nu} &= (\partial_\mu - igA_\mu)A_\nu - (\partial_\nu - igA_\nu)A_\mu \\ &= \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]. \end{aligned} \quad (12.20)$$

The commutator term is the essential non-Abelian correction. It vanishes for $U(1)$, where all Lie-algebra elements commute, and Maxwell's field tensor is recovered.

12.7.2 Curvature as a commutator of covariant derivatives

The same tensor follows intrinsically from the commutator $[D_\mu, D_\nu]$. Acting on an arbitrary multiplet Φ ,

$$\begin{aligned} [D_\mu, D_\nu]\Phi &= [\partial_\mu - igA_\mu, \partial_\nu - igA_\nu]\Phi \\ &= -ig(\partial_\mu A_\nu)\Phi + ig(\partial_\nu A_\mu)\Phi - g^2[A_\mu, A_\nu]\Phi. \end{aligned}$$

The terms $A_\nu\partial_\mu\Phi$ and $A_\mu\partial_\nu\Phi$ cancel pairwise after the product rule is applied. Factoring out $-ig$ gives

$$[D_\mu, D_\nu]\Phi = -ig(\partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu])\Phi.$$

Hence

$$F_{\mu\nu} = \frac{i}{g}[D_\mu, D_\nu]. \quad (12.21)$$

This formula explains the geometric meaning of $F_{\mu\nu}$: it measures the failure of covariant derivatives in two spacetime directions to commute, so it is the curvature of the connection.

12.7.3 Gauge covariance of the field strength

Use the operator law (12.16):

$$\begin{aligned} F'_{\mu\nu} &= \frac{i}{g}[D'_\mu, D'_\nu] \\ &= \frac{i}{g}[RD_\mu R^{-1}, RD_\nu R^{-1}] \\ &= \frac{i}{g}R[D_\mu, D_\nu]R^{-1}. \end{aligned}$$

Therefore

$$\boxed{F'_{\mu\nu} = RF_{\mu\nu}R^{-1}.} \quad (12.22)$$

The field strength is not generally unchanged; it transforms *covariantly* by conjugation. This is sufficient to construct a gauge invariant scalar by taking a trace.

Because $\partial_\mu A_\nu$ lies in the same vector space as A_ν , and because a Lie algebra is closed under commutators,

$$F_{\mu\nu} \in \mathfrak{so}(n)$$

in the real convention, or in its Hermitian-generator realization used above. If

$$[H^a, H^b] = if^{ab}_c H^c,$$

then

$$\begin{aligned} F_{\mu\nu} &= \sum_c F_{\mu\nu}^c H^c, \\ F_{\mu\nu}^c &= \partial_\mu A_\nu^c - \partial_\nu A_\mu^c + gf^{ab}_c A_\mu^a A_\nu^b. \end{aligned} \quad (12.23)$$

The final quadratic term is absent in an Abelian theory and is responsible for gauge-boson self-interactions.

12.8 The Yang–Mills Lagrangian

12.8.1 Trace form and proof of gauge invariance

The gauge-field Lagrangian proposed in the lecture is

$$\mathcal{L}_{\text{YM}} = -\frac{1}{2} \text{tr}(F_{\mu\nu} F^{\mu\nu}). \quad (12.24)$$

Under a gauge transformation,

$$\begin{aligned} \text{tr}(F'_{\mu\nu} F'^{\mu\nu}) &= \text{tr}(RF_{\mu\nu}R^{-1}RF^{\mu\nu}R^{-1}) \\ &= \text{tr}(RF_{\mu\nu}F^{\mu\nu}R^{-1}) \\ &= \text{tr}(F_{\mu\nu}F^{\mu\nu}), \end{aligned}$$

where cyclicity of the trace was used in the last line. Thus $\mathcal{L}_{\text{YM}}$ is gauge invariant.

Choose the generator normalization written in the notes,

$$\mathrm{tr}(H^a H^b) = \frac{1}{2} \delta^{ab}. \quad (12.25)$$

Then

$$\begin{aligned} \mathcal{L}_{\mathrm{YM}} &= -\frac{1}{2} \sum_{\mu, \nu} \sum_{a, b} F_{\mu\nu}^a F^{\mu\nu b} \mathrm{tr}(H^a H^b) \\ &= -\frac{1}{4} \sum_{\mu, \nu} \sum_a F_{\mu\nu}^a F^{\mu\nu a}. \end{aligned}$$

Since $F_{\nu\mu} = -F_{\mu\nu}$, summing over all ordered pairs (μ, ν) counts every independent pair twice. Equivalently,

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{2} \sum_{\mu < \nu} \sum_a F_{\mu\nu}^a F^{\mu\nu a}. \quad (12.26)$$

The factor $1/4$ in the usual formula is the familiar free-field normalization together with this antisymmetric double counting.

12.8.2 Self-interactions of non-Abelian gauge particles

Substituting (12.20) into (12.24) shows three types of terms:

$$\mathcal{L}_{\mathrm{YM}} \sim (\partial A)^2 + g(\partial A)A^2 + g^2 A^4.$$

The first is the gauge-field kinetic term. The second and third are, respectively, cubic and quartic gauge-field interactions. They exist because $[A_\mu, A_\nu] \neq 0$. Thus, unlike the photon in pure Abelian electromagnetism, non-Abelian gauge particles carry the gauge charge and interact with one another.

12.9 The complete gauge-invariant theory

Combining the matter, interaction, and gauge-field pieces gives

$$\mathcal{L} = \mathcal{L}_{\mathrm{free}} + \mathcal{L}_{\mathrm{int}} + \mathcal{L}_{\mathrm{YM}}.$$

In compact covariant form,

$$\mathcal{L} = \frac{1}{2} (D_\mu \Phi)^T D^\mu \Phi - \frac{1}{2} m^2 \Phi^T \Phi - \frac{1}{2} \mathrm{tr}(F_{\mu\nu} F^{\mu\nu}). \quad (12.27)$$

Writing out every term gives the final expression displayed in the handwritten lecture:

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} \partial_\mu \Phi^T \partial^\mu \Phi - \frac{1}{2} m^2 \Phi^T \Phi \\ &\quad + \frac{1}{2} [-ig \Phi^T A_\mu^T \partial^\mu \Phi - ig \partial_\mu \Phi^T A^\mu \Phi - g^2 \Phi^T A_\mu^T A^\mu \Phi] \\ &\quad - \frac{1}{2} \mathrm{tr}[(\partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu])(\partial^\mu A^\nu - \partial^\nu A^\mu - ig[A^\mu, A^\nu])]. \end{aligned} \quad (12.28)$$

Every term is Lorentz invariant. The three pieces are not separately arbitrary: their precise combination and their relative interaction coefficients are fixed by local $SO(n)$ gauge invariance.

What to retain from this chapter

1. The free multiplet $\Phi = (\varphi_1, \dots, \varphi_n)^T$ obeys $(\square + m^2)\Phi = 0$ and has global $SO(n)$ internal symmetry.
2. Infinitesimal $SO(n)$ generators are skew-symmetric matrices:

$$\mathfrak{so}(n) = \{T : T^T = -T\}, \quad \dim \mathfrak{so}(n) = \frac{1}{2}n(n-1).$$

3. Global internal symmetry gives one conserved Noether current $j^{\mu a} = \partial^\mu \Phi^T T^a \Phi$ for every generator.
4. Allowing $R = R(x)$ produces the infinite-dimensional gauge group $C^\infty(M, SO(n))$, but the ordinary derivative of Φ then fails to transform covariantly.
5. The covariant derivative and connection transform as

$$D_\mu = \partial_\mu - igA_\mu, \quad D'_\mu = RD_\mu R^{-1},$$

$$A'_\mu = RA_\mu R^{-1} + \frac{i}{g}R\partial_\mu R^{-1}.$$

6. The curvature is

$$F_{\mu\nu} = \frac{i}{g}[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu],$$

and transforms covariantly: $F'_{\mu\nu} = RF_{\mu\nu}R^{-1}$.

7. The trace removes the conjugation, so $\mathcal{L}_{\text{YM}} = -\frac{1}{2} \text{tr}(F_{\mu\nu}F^{\mu\nu})$ is gauge invariant.
8. Minimal coupling generates matter–gauge interactions, while the commutator in $F_{\mu\nu}$ generates cubic and quartic self-interactions of non-Abelian gauge fields.